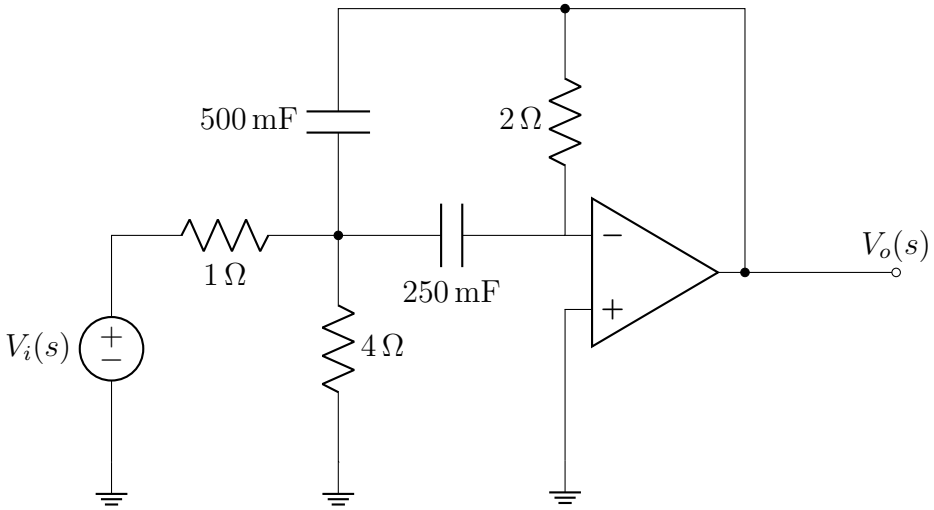
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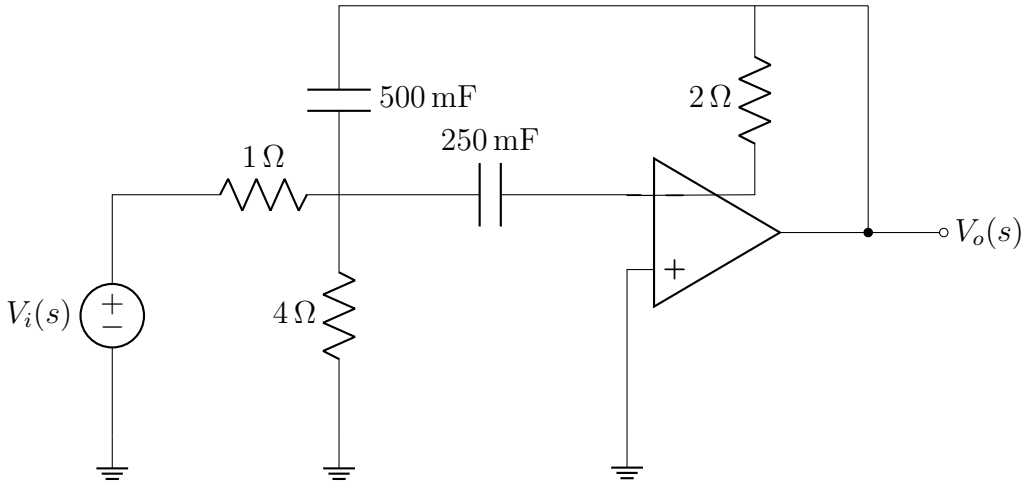
1. Suppose we are designing a low-pass Butterworth filter with passband gain between 0 dB and -2 dB for $0 \leq \omega \leq 100$ rad/s, and stopband gain no greater than -20 dB for $\omega \geq 400$ rad/s.
   (a) Find the filter order n and the -3 dB frequency ω_c.
   (b) Find the normalized transfer function $\tilde{H}(s)$. Leave it in factored form if convenient.
   (c) Find the transfer function $H(s)$ for the designed filter.
   (d) You have two op-amps, 10 kΩ resistors, and capacitors no larger than 100 μF. Design a circuit to implement the filter using the Sallen–Key topology below. Use $R_1 = R_2 = 10\,\text{k}\Omega$. What values do you choose for C_1 and C_2?



2. (a) A multiple-feedback active filter has transfer function

$$H(s) = \frac{2\zeta\omega_0 K s}{s^2 + 2\zeta\omega_0 s + \omega_0^2}.$$

For the circuit below, find K, ζ, and ω_0.



(b) Design a low-pass Butterworth filter with gain -2 dB at 1 kHz and gain -20 dB at 4 kHz. Find the order n and cutoff frequency ω_c in rad/s. Be careful to convert the frequencies to radians per second.

(c) Find the transfer function $H(s)$ for this Butterworth filter.

Answer Key

1(a) $n = 2, \quad \omega_c \approx 120.42\,\text{rad/s}$

1(b) $\tilde{H}(s) = \dfrac{1}{s^2 + \sqrt{2}s + 1}$

1(c) $H(s) = \dfrac{\omega_c^2}{s^2 + \sqrt{2}\,\omega_c s + \omega_c^2}, \quad \omega_c \approx 120.42\,\text{rad/s}$

1(d) $C_1 \approx 1.17\,\mu\text{F}, \quad C_2 \approx 0.587\,\mu\text{F}$

2(a) $K = -\frac{2}{3}, \quad \zeta = \frac{3}{2\sqrt{5}}, \quad \omega_0 = \sqrt{5}\,\text{rad/s}$

2(b) $n = 2, \quad \omega_c \approx 7566.08\,\text{rad/s}$

2(c) $H(s) = \dfrac{\omega_c^2}{s^2 + \sqrt{2}\,\omega_c s + \omega_c^2}, \quad \omega_c \approx 7566.08\,\text{rad/s}$